

Vector Norms

ℓ_p Norm: $\|x\|_p = (\sum_{i=1}^d |x_i|^p)^{1/p}$
Infinity Norm: $\|x\|_\infty = \max_i |x_i|$

Proof $\|x\|_2 \leq \|x\|_1$
 $\|x\|_1^2 = (\sum_i |x_i|)^2 = \sum_i |x_i|^2 + \sum_{i \neq j} |x_i| |x_j|$
Cross-product terms ≥ 0 , so $\|x\|_1^2 \geq \|x\|_2^2$, hence $\|x\|_1 \geq \|x\|_2$.

Proof $\|x\|_\infty = \lim_{p \rightarrow \infty} \|x\|_p$
Let $M = \|x\|_\infty = \max_i |x_i|$. Since $M^p \leq \sum_i |x_i|^p$ (one term), taking p -th roots: $M \leq \|x\|_p$. Since $|x_i| \leq M$ for all i : $\sum_i |x_i|^p \leq dM^p$, so $\|x\|_p \leq d^{1/p} M$. Combining: $M \leq \|x\|_p \leq d^{1/p} M$. As $p \rightarrow \infty$, $d^{1/p} \rightarrow 1$; by the Squeeze Theorem $\lim_{p \rightarrow \infty} \|x\|_p = M$.

Weak and Strong Law of Large Numbers
 $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$, $\mathbb{E}[\bar{x}_n] = \mu$, $\text{Var}(\bar{x}_n) = \sigma^2/n$
Chebyshev: $P(|Y - \mathbb{E}[Y]| > \epsilon) \leq \text{Var}(Y)/\epsilon^2$

Proof of WLLN (via Chebyshev)
Apply Chebyshev with $Y = \bar{x}_n$:

$$P(|\bar{x}_n - \mu| > \epsilon) \leq \frac{\text{Var}(\bar{x}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \xrightarrow{n \rightarrow \infty} 0$$

Hence $\bar{x}_n \xrightarrow{p} \mu$. □

WLLN vs SLLN
WLLN: for fixed $\epsilon > 0$, $P(|\bar{x}_n - \mu| > \epsilon) \rightarrow 0$ (convergence in probability at each n). SLLN: $P(\lim_{n \rightarrow \infty} \bar{x}_n = \mu) = 1$ (almost-sure convergence; the entire trajectory stays close to μ eventually).

High-Dimensional Distance Concentration
Std-Dev Approximation via Delta Method

$$S_d = \sum_{j=1}^d z_j, \bar{z}_d = S_d/d, D_d = S_d^{1/p} = d^{1/p} \bar{z}_d^{1/p}$$
$$g(s) = s^{1/p}, g'(s) = \frac{1}{p} s^{1/p-1}. \text{ Delta Method:}$$

$$\text{Var}(g(\bar{z}_d)) \approx \frac{\sigma_{\bar{z}}^2}{d} (g'(\mu_z))^2 = \frac{\sigma_z^2}{d p^2} \mu_z^{2/p-2}$$

$$\text{Var}(D_d) = d^{2/p} \text{Var}(g(\bar{z}_d)) \approx \frac{\sigma_z^2}{p^2} \mu_z^{2/p-2} d^{2/p-1}$$

$$\text{std}(D_d) \approx \frac{\sigma_z}{p} \mu_z^{1/p-1} d^{1/p-1/2}$$

Concentration Ratio
 $\mathbb{E}[D_d] \approx (d\mu_z)^{1/p} = d^{1/p} \mu_z^{1/p}$ (LLN).

$$\frac{\mathbb{E}[D_d]}{\text{std}(D_d)} \approx \frac{d^{1/p} \mu_z^{1/p}}{\frac{\sigma_z}{p} \mu_z^{1/p-1} d^{1/p-1/2}} = \frac{p\mu_z}{\sigma_z} \sqrt{d} = O(\sqrt{d})$$

ℓ_1 ($p=1$) preserves more contrast than ℓ_2 ($p=2$) in high dimensions.

Vectorised Distance Computation

$$\|x_i - q\|_2^2 = x_i^\top x_i - 2x_i^\top q + q^\top q$$
$$d^2 = \text{diag}(X X^\top) - 2Xq + \|q\|_2^2 \mathbf{1}_n$$

Proof: $[\text{diag}(X X^\top)]_i = x_i^\top x_i$; $[Xq]_i = x_i^\top q$; third term is scalar broadcast.
Gains: BLAS/LAPACK routines, SIMD parallelism, cache-contiguous memory access.

Decision Trees: Approximation and Stability

Diagonal Boundary Approximation

CART uses axis-aligned splits $x_j \leq t$. A depth- d tree has 2^d leaves; the staircase approximation to the diagonal $x_2 = x_1$ has error $\epsilon \sim 2^{-d}$. Solving $2^d \sim 1/\epsilon$ gives $d = O(\log(1/\epsilon))$.

Statistical Consistency

Thm (Tree Consistency). Assume $y = f^*(x) + \epsilon$ with $\mathbb{E}[\epsilon|x] = 0$, $\text{Var}(\epsilon|x) = \sigma^2$, f^* bounded and continuous; leaves satisfy $\max_t \text{diam}(R_t^{(n)}) \xrightarrow{a.s.} 0$ and $\min_t |R_t^{(n)}| \xrightarrow{a.s.} \infty$. Then $\mathbb{E}[(f^{(n)}(x_0) - f^*(x_0))^2] \rightarrow 0$.

Proof. Let $R^{(n)} = R^{(n)}(x_0)$ be the random leaf containing x_0 and $f^{(n)}(x_0) = \frac{1}{|R^{(n)}|} \sum_{i:x_i \in R^{(n)}} y_i$. Condition on $R^{(n)}$:

$$\mathbb{E}[(f^{(n)}(x_0) - f^*(x_0))^2] = \mathbb{E}[\text{Var}(f^{(n)} | R^{(n)}) + (\mathbb{E}[f^{(n)} | R^{(n)}] - f^*(x_0))^2]$$

$$\text{Bias: } \mathbb{E}[f^{(n)} | R^{(n)}] = \frac{1}{|R^{(n)}|} \sum_{i:x_i \in R^{(n)}} f^*(x_i) \xrightarrow{a.s.} f^*(x_0)$$

$f^*(x_0)$ as $\text{diam} \rightarrow 0$ and f^* continuous. Bounded f^* gives $\mathbb{E}[\text{bias}^2] \rightarrow 0$ by DCT.

Variance: $\text{Var}(f^{(n)} | R^{(n)}) = \sigma^2/|R^{(n)}| \xrightarrow{a.s.} 0$ since $|R^{(n)}| \rightarrow \infty$; DCT gives $\mathbb{E}[\text{var}] \rightarrow 0$.

Fixed-depth trees: leaves do not shrink $\Rightarrow$ bias does not vanish $\Rightarrow$ not consistent.

Instability: Greedy splitting means a small perturbation in one point can alter the root split, cascading to change all downstream splits.

Bias-Variance Decomposition

Lemma (General). For random estimator z and fixed target z_0 :

$$\mathbb{E}[(z - z_0)^2] = \underbrace{\text{Var}(z)}_{\text{variance}} + \underbrace{(\mathbb{E}[z] - z_0)^2}_{\text{bias}^2}$$

Proof. Write $z - z_0 = (z - \mathbb{E}[z]) + (\mathbb{E}[z] - z_0)$ and square. Cross term: $2(\mathbb{E}[z] - z_0) \mathbb{E}[z - \mathbb{E}[z]] = 0$. □

Thm (Regression B-V). With $y = f^*(x) + \epsilon$, $\mathbb{E}[\epsilon|x] = 0$, $\text{Var}(\epsilon|x) = \sigma^2$, $f(x) = \mathbb{E}_D[f(x; D)]$: $\bar{E}^* = \underbrace{\mathbb{E}_x[(f(x) - f^*(x))^2]}_{\text{Bias}^2} + \underbrace{\mathbb{E}_x[\mathbb{E}_D[(f(x; D) - f(x))^2]]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}}$

Proof. $\bar{E}^* = \mathbb{E}_D[\mathbb{E}_x[\epsilon[(f(x; D) - f^*(x) - \epsilon)^2]]]$. Expand over ϵ : $\mathbb{E}_\epsilon[(f - f^* - \epsilon)^2] = (f - f^*)^2 + \sigma^2$ (since $\mathbb{E}[\epsilon|x] = 0$). Apply the general Lemma pointwise with $z = f(x; D)$, $z_0 = f^*(x)$, then integrate over x . □

Conditional Decomposition for Regression Trees

Let $h = f^{(n)}(x_0)$, $h^* = f^*(x_0)$, $R = R^{(n)}(x_0)$.

By LTE: $\mathbb{E}[(h - h^*)^2] = \mathbb{E}[\mathbb{E}[(h - h^*)^2 | R]]$

Apply Lemma conditional on R :

$$\mathbb{E}[(h - h^*)^2] = \mathbb{E}[\text{Var}(h | R) + (\mathbb{E}[h | R] - h^*)^2]$$

Entropy and Information Gain

$$H(y) = -\sum_c p_c \ln p_c, \quad H(y|s) = \frac{|D_l|}{|D|} H(D_l) + \frac{|D_r|}{|D|} H(D_r)$$

$IG = H(y) - H(y|s)$

Example: Parent (6P, 4N), Left (4P, 1N), Right (2P, 3N):

$H_{\text{par}} \approx 0.673$, $H(D_l) \approx 0.500$, $H(D_r) \approx 0.673$

$H(y|s) = 0.5(0.500) + 0.5(0.673) = 0.587$, $IG = 0.086 > 0$

Gini Index (Probabilistic Interpretation)

$\text{cost}(R_t) = 1 - \sum_c (\pi_c^t)^2 = P(\text{two i.i.d. draws from } R_t \text{ differ})$.

Proof. $P(\text{same class}) = \sum_{c=1}^C P(y_1 = c)P(y_2 = c) = \sum_c (\pi_c^t)^2$. Hence $P(\text{different}) = 1 - \sum_c (\pi_c^t)^2$. □

Cost-Complexity Pruning

Objective: $L_\alpha(T) = R_n(T) + \alpha|T|$,

where $R_n(T) = \frac{1}{n} \sum_i \mathbf{1}\{f_T(x_i) \neq y_i\}$.

Proof $R_n(T) = 0$ for fully-grown tree

Each leaf contains exactly one point (x_i, y_i) , so $f_T(x_i) = y_i$ for all i , giving $R_n(T) = 0$. This corresponds to memorisation: high variance, poor generalisation.

Proof $\alpha > 0$ yields single-leaf tree

For $|T| > 1$: $L_\alpha(T) \geq \alpha|T| \geq 2\alpha$. For T_{root} : $L_\alpha(T_{\text{root}}) = R_n(T_{\text{root}}) + \alpha$. As $\alpha \rightarrow \infty$ the penalty $\alpha|T|$ dominates, so the minimiser has $|T| = 1$.

Bias-Variance via α : large $\alpha \Rightarrow$ smaller tree $\Rightarrow$ variance $\downarrow$, bias $\uparrow$; optimal generalisation at intermediate α .

Bayesian Linear Regression

Model: $y = X\theta + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$. Prior: $\theta \sim \mathcal{N}(\mu_0, \Sigma_0)$.

Posterior: $p(\theta|X, y) = \mathcal{N}(\mu_N, \Sigma_N)$ where

$$\Sigma_N^{-1} = \frac{1}{\sigma^2} X^\top X + \Sigma_0^{-1}, \quad \mu_N = \Sigma_N \left(\frac{1}{\sigma^2} X^\top y + \Sigma_0^{-1} \mu_0 \right)$$

Proof (sketch). Posterior $\propto$ likelihood $\times$ prior. Both are Gaussian, so the log-posterior is a quadratic in θ . Expand and collect: quadratic term gives Σ_N^{-1} ; linear term gives $\Sigma_N^{-1} \mu_N$. Complete the square to read off $\mathcal{N}(\mu_N, \Sigma_N)$. □

Regime $\tau^2 \rightarrow \infty$ (flat prior), $\Sigma_0 = \tau^2 I$:

$$\Sigma_N^{-1} \rightarrow \sigma^{-2} X^\top X, \quad \mu_N \rightarrow (X^\top X)^{-1} X^\top y = \hat{\theta}_{\text{OLS}}$$

Regime $\tau^2 \rightarrow 0$ (infinite precision prior):

$\Sigma_N^{-1} \rightarrow \infty \cdot I_p \Rightarrow \Sigma_N \rightarrow 0$, $\mu_N \rightarrow 0$; prior dominates, posterior collapses to $\theta = 0$.

MAP = Ridge With $\Sigma_0 = \frac{\sigma^2}{n\lambda} I$:

$$\log p(\theta|X, y) \propto -\frac{1}{2\sigma^2} (\|y - X\theta\|_2^2 + n\lambda\|\theta\|_2^2)$$

$$\Rightarrow \hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \frac{1}{n} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 = \hat{\theta}_{\text{ridge}}$$

Posterior Predictive For new x_* with $y_* = x_*^\top \theta + \epsilon_*$, $\epsilon_* \sim \mathcal{N}(0, \sigma^2)$:

$$y_* | x_*, X, y \sim \mathcal{N}(\underbrace{(\underbrace{x_*^\top \mu_N}_{\text{aleatoric}} + \underbrace{\Sigma_N x_*}_{\text{epistemic}}), \sigma^2}$$

Proof. $x_*^\top \theta | X, y \sim \mathcal{N}(x_*^\top \mu_N, x_*^\top \Sigma_N x_*)$ by affine transform. Add independent ϵ_* ; variances sum. □

Bayesian Linear Regression (Unknown Variance)

Prior: $\theta | \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2 \Sigma_0)$, $\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$.

Posterior: $\theta | \sigma^2, X, y \sim \mathcal{N}(\mu_N, \sigma^2 \Sigma_N)$, $\sigma^2 | X, y \sim \text{Inv-Gamma}(\alpha_N, \beta_N)$

$$\Sigma_N^{-1} = X^\top X + \Sigma_0^{-1}, \quad \mu_N = \Sigma_N (X^\top y + \Sigma_0^{-1} \mu_0)$$

$$\alpha_N = \alpha_0 + n/2$$

$$\beta_N = \beta_0 + \frac{1}{2} (y^\top y + \mu_0^\top \Sigma_0^{-1} \mu_0 - \mu_N^\top \Sigma_N^{-1} \mu_N)$$

Logistic Regression

Logistic Loss from MLE

Model: $p(y = 1|x; \theta) = \sigma(\theta^\top x)$, $y \in \{-1, +1\}$. Independence gives:

$$p(y|x; \theta) = \sigma(y\theta^\top x) = \frac{e^{y\theta^\top x}}{1 + e^{y\theta^\top x}}$$

$$\Rightarrow \ell = -\ln p(y|x; \theta) = \ln(1 + e^{-y\theta^\top x})$$

$$\text{Empirical risk: } L(\theta) = \frac{1}{n} \sum_i \ln(1 + e^{-y_i \theta^\top x_i})$$

Derivative of Sigmoid

$$\sigma(z) = (1 + e^{-z})^{-1}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z))$$

Gradient of Logistic Loss

$$\nabla_{\theta} L(\theta) = \frac{1}{n} \sum_i (h_i(\theta) - y_i) x_i = \frac{1}{n} X^\top (h(\theta) - y)$$

Proof. Let $z_i = \theta^\top x_i$, $h_i = \sigma(z_i)$.

$$\nabla_{\theta} L = -\frac{1}{n} \sum_i \left[\frac{y_i}{h_i} \cdot h_i(1 - h_i) - \frac{1 - y_i}{1 - h_i} \cdot h_i(1 - h_i) \right] x_i$$

$$= -\frac{1}{n} \sum_i [y_i(1 - h_i) - (1 - y_i)h_i] x_i = \frac{1}{n} \sum_i (h_i - y_i) x_i$$

Linear Decision Boundary

Predict 1 iff $\sigma(\theta^\top x) \geq \frac{1}{2}$ iff $\theta^\top x \geq 0$; boundary is hyperplane $\theta^\top x = 0$. □

Softmax (Multiclass)

$$\text{Binary: } p(y = 1|x) = \frac{e^{\theta_1^\top x}}{e^{\theta_1^\top x} + e^{\theta_2^\top x}}$$

$$\text{Generalises to } p(y = c|x; \theta) = \frac{e^{\theta_c^\top x}}{\sum_{c'=1}^C e^{\theta_{c'}^\top x}}$$

Cross-Entropy Decomposition

$$H(q, p) = -\sum_c q(c) \ln p(c) = -\underbrace{\sum_c q(c) \ln q(c)}_{H(q)} + \underbrace{\sum_c q(c) \ln \frac{q(c)}{p(c)}}_{\text{KL}(q||p) \geq 0}$$

Hence minimising cross-entropy $\equiv$ minimising $\text{KL}(q||p)$; equality iff $p = q$. □

Perfect Separation Paradox

Loss: $L(\theta) = \frac{1}{n} \sum_i \ln(1 + e^{-y_i \theta^\top x_i})$

If $\exists \theta^*$: $y_i \theta^{*\top} x_i > 0 \forall i$, scale $\theta = \alpha \theta^*$:

$L(\alpha \theta^*) \rightarrow 0$ as $\alpha \rightarrow \infty$, but $L(\theta) > 0 \forall \theta \in \mathbb{R}^d$, so inf is unattained $\Rightarrow$ MLE does not exist.

Fix: ℓ_2 regularisation. $J(\theta) = L(\theta) + \lambda \|\theta\|_2^2$. As $\|\theta\| \rightarrow \infty$, $\|\theta\|_2^2 \rightarrow \infty$, so J is coercive and a finite minimiser exists.

Loss Functions: Decision-Theoretic Foundations

Bayes-Optimal Predictor (Pointwise)

$f^*(x) \in \arg \min_{t \in \mathbb{R}} \mathbb{E}_{p^*}(y|x) \ell(y, t)$

Proof. By tower rule: $\mathbb{E}_{p^*}(x, y) [\ell(y, f(x))] = \mathbb{E}_x [\mathbb{E}_y [y|x] \ell(y, f(x))]$. Minimise inner expectation pointwise at each x . □

Squared Loss $\rightarrow$ Conditional Mean

$$\ell(y, t) = (y - t)^2; \quad \frac{\partial}{\partial t} \mathbb{E}[(y - t)^2 | x] = -2\mathbb{E}[y|x] + 2t = 0 \Rightarrow t^* = \mathbb{E}[y|x]$$

Absolute Loss $\rightarrow$ Conditional Median

$$h(t) = \mathbb{E}[|y - t| | x] = \int_{-\infty}^t -\infty(t - y) p(y|x) dy + \int_{\infty}^t (y - t) p(y|x) dy$$

Leibniz rule: $h'(t) = P(y \leq t|x) - P(y > t|x) = 2P(y \leq t|x) - 1$

$h'(t^*) = 0 \Rightarrow P(y \leq t^*|x) = \frac{1}{2} \Rightarrow t^* = \text{median}(y|x)$

NLL is Proper

$\mathbb{E}_{p^*}[-\ln p_f(y|x)] = H(p^*) + \text{KL}(p^*||p_f) \geq H(p^*)$, with equality iff $p_f = p^*$.

Proof. Add/subtract $\ln p^*$ inside the expectation. □

0-1 Loss: Bayes Classifier and Impropriety

$f^*(x) = \text{sign}(\eta(x) - \frac{1}{2})$; error = $\min\{\eta(x), 1 - \eta(x)\}$. Not proper: f^* depends only on the sign of $\eta - \frac{1}{2}$, so distinct η values map to the same prediction and η cannot be recovered.

Cross-Entropy Strictly Proper

Risk $\eta(-\ln t + (1 - \eta)(-\ln(1 - t)))$; FOC: $-\eta/t + (1 - \eta)/(1 - t) = 0 \Rightarrow t^* = \eta(x)$. Strictly convex on $(0, 1) \Rightarrow$ unique minimiser recovers the full conditional distribution. □

Logistic Loss Strictly Proper

$\phi(t) = \eta \ln(1 + e^{-t}) + (1 - \eta) \ln(1 + e^t)$; $\phi'(t) = 0 \Rightarrow e^t = \eta/(1 - \eta) \Rightarrow f^*(x) = \ln \frac{\eta(x)}{1 - \eta(x)}$ (log-odds). Log-odds map invertible in $\eta \Rightarrow$ strictly proper. □

Likelihood-Based Loss Functions

$y_i, \theta \Rightarrow -\ln p(y_i|x; \theta)$. Gaussian $\Rightarrow$ squared loss; Laplace $\Rightarrow$ absolute loss.

Gaussian MLE = Least Squares

$p(y|X; \theta) = \prod_i \mathcal{N}(y_i; \theta^\top x_i, \sigma^2)$. Log-likelihood:

$$\ln p = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - \theta^\top x_i)^2$$

Maximising over $\theta \equiv$ minimising $\sum_i (y_i - \theta^\top x_i)^2$. □

Poisson Regression NLL $y_i | x_i \sim \text{Pois}(\lambda_i)$, $\lambda_i = e^{\theta^\top x_i}$:

$$\text{NLL} = \sum_i (e^{\theta^\top x_i} - y_i \theta^\top x_i) + \text{const}$$

Heteroskedastic Gaussian $y|x; \theta \sim \mathcal{N}(f(x; \theta), \exp(h(x; \theta)))$:

$$\ell \propto \frac{1}{2} h(x; \theta) + \frac{(y - f(x; \theta))^2}{2 \exp(h(x; \theta))}$$

High variance: error down-weighted. $\frac{1}{2} h$ term prevents $h \rightarrow \infty$ trivially.

Normal Equations and Ridge

OLS: $\nabla_{\theta} \frac{1}{2} \|X\theta - y\|_2^2 = X^\top (X\theta - y) = 0 \Rightarrow X^\top X \hat{\theta} = X^\top y$

If $X^\top X$ invertible: $\hat{\theta} = (X^\top X)^{-1} X^\top y$.

Ridge: $\nabla(\frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2) = 0 \Rightarrow (X^\top X + n\lambda I) \hat{\theta} = X^\top y$

$\lambda > 0$ makes $X^\top X + n\lambda I$ PD (hence invertible): $\hat{\theta} = (X^\top X + n\lambda I)^{-1} X^\top y$. □

Ridge Bias and Variance

$S_\lambda = (X^\top X + \lambda I)^{-1} X^\top X$. $\hat{\theta}_\lambda = S_\lambda \theta^* + (X^\top X + \lambda I)^{-1} X^\top \epsilon$. $\mathbb{E}[\hat{\theta}_\lambda] = S_\lambda \theta^*$; Bias($\hat{\theta}_\lambda$) = $(S_\lambda - I) \theta^*$ (nonzero for $\lambda > 0$).

Cov($\hat{\theta}_\lambda$) = $\sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}$ (shrinks as $\lambda \uparrow$). □

Multivariate Normal: Key Properties

Zero Cov $\Rightarrow$ Independence (Gaussian case only)

Cov(x_1, x_2) = 0 <

PSD Kernel Examples

Polynomial: $k(x, x') = (1 + x x')^2 = \phi(x)^\top \phi(x')$ with $\phi(x) = (1, \sqrt{2}x, x^2)^\top$. $v^\top K v = \sum_i v_i (1 + 2x_i x_j + x_i^2 x_j^2) \geq 0$.

Exponential: $k(x, x') = e^{x x'} = \sum_{m=0}^\infty \frac{x^m}{\sqrt{m!}} \frac{x'^m}{\sqrt{m!}}$. $v^\top K v = \sum_{m=0}^\infty \frac{1}{m!} (\sum_i v_i x_i^m)^2 \geq 0$. □

KRR Bias and Variance

Smoother: $f = Sy, S = K(K + n\lambda I)^{-1}$ (symmetric).
 $\mathbb{E}[f] = Sf^*$; Bias = $(S - I)f^* = -n\lambda(K + n\lambda I)^{-1}f^*$.
 $\text{Cov}(f) = \sigma^2 S^2$; $\mathbb{E}\|f - f^*\|_2^2 = \|(S - I)f^*\|_2^2 + \sigma^2 \text{tr}(S^2)$.
Spectral view: $K = U\Lambda U^\top \Rightarrow S = \sum_i \frac{\mu_i}{\mu_i + n\lambda} u_i u_i^\top$; large μ_i (smooth modes) preserved, small μ_i (noisy) suppressed. □

Support Vector Regression (SVR)

$\ell_\epsilon(r) = \max(0, |r| - \epsilon)$; as $\epsilon \rightarrow 0$, reduces to KLD ($|r|$).
Support vectors: points with $|y_i - \theta^\top \phi(x_i)| \geq \epsilon$ (outside the tube).
Robustness: Influence function $\psi(r) \in \partial \ell(r)$: KRR $\psi = 2r$ (unbounded); KLD/SVR $|\psi| \leq 1$ (bounded) $\Rightarrow$ robust to outliers.

SVR Dual Primal $\min_\theta, \xi \frac{1}{n} \sum_i (\xi_i^+ + \xi_i^-) + \lambda \|\theta\|_2^2$ s.t. slack constraints. Stationarity in θ : $\hat{\theta} = \frac{1}{2\lambda} \sum_i \alpha_i \phi(x_i)$, $\alpha_i = \alpha_i^+ - \alpha_i^-$. Dual (after rescaling): $\min_\alpha \frac{1}{2} \alpha^\top K \alpha - \alpha^\top y + \epsilon \|\alpha\|_1$ s.t. $|\alpha_i| \leq \frac{1}{2\lambda n}$. □

SVC (Soft-Margin SVM) Dual $\theta = \frac{1}{2\lambda} \sum_i \alpha_i y_i \phi(x_i)$. Dual: $\min_\alpha \frac{1}{2} \alpha^\top (YKY) \alpha - 1^\top \alpha$ s.t. $0 \leq \alpha_i \leq \frac{1}{2\lambda n}$, $Y = \text{diag}(y)$ □

Weak Duality For $\min_\omega f(\omega)$ s.t. $g_i(\omega) \leq 0$, dual $g(\alpha) = \min_\omega L(\omega, \alpha)$: $\max_{\alpha \geq 0} g(\alpha) \leq \min_{g_i \leq 0} f(\omega)$

Proof. Fix $\alpha \geq 0$ and feasible ω : $\sum_i \alpha_i g_i(\omega) \leq 0 \Rightarrow L(\omega, \alpha) \leq f(\omega) \Rightarrow g(\alpha) \leq f(\omega)$ for all feasible ω . □

Gaussian Processes

Construction and Existence A GP $f \sim \mathcal{GP}(\mu, \kappa)$ exists iff κ is PSD (Kolmogorov extension theorem guarantees consistent marginals).
Linear model $f(x) = \phi(x)^\top \theta$, $\theta \sim \mathcal{N}(\mu_\theta, I)$ is a GP with $\kappa(x, x') = \phi(x)^\top \phi(x')$.

GP Regression Posterior

Prior: $f \sim \mathcal{N}(\mu, K)$, noise: $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $y = f + \varepsilon \sim \mathcal{N}(\mu, K + \sigma^2 I)$.

Joint: $\begin{bmatrix} y \\ y_* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_* \\ \mu_* \end{bmatrix}, \begin{bmatrix} K + \sigma^2 I & k_* \\ k_*^\top & \kappa_{**} \end{bmatrix}\right)$
Gaussian conditioning ($b|a \sim \mathcal{N}(\mu_b + C^\top A^{-1}(a - \mu_a), B - C^\top A^{-1}C)$):

$f_*|y \sim \mathcal{N}(\mu_n(x_*), \sigma_n^2(x_*))$
 $\mu_n(x_*) = \mu_* + k_*^\top (K + \sigma^2 I)^{-1} (y - \mu)$
 $\sigma_n^2(x_*) = \kappa_{**} - k_*^\top (K + \sigma^2 I)^{-1} k_*$ □

Posterior Variance Monotone in Data
 $\sigma_n^2(x) = \kappa(x, x) - k^\top (K + \sigma^2 I)^{-1} k$. Enlarging the dataset grows $k^\top (K + \sigma^2 I)^{-1} k$ (extra orthogonal directions of explanation), so $\sigma_n^2(x)$ cannot increase. □

Log Marginal Likelihood: $\ln p(y) = -\frac{1}{2} (y - \mu)^\top (K + \sigma^2 I)^{-1} (y - \mu) - \frac{1}{2} \ln |K + \sigma^2 I| - \frac{n}{2} \ln(2\pi)$
Used to optimise kernel hyperparameters (data-fit vs log-det complexity penalty).

GP for Classification

Latent function $f(\cdot)$: $f \sim \mathcal{N}(0, K)$, $K_{ij} = k(x_i, x_j)$.
Likelihood: $p(y_i|f_i) = \sigma(y_i f_i)$. Posterior $p(f|y) \propto \mathcal{N}(f|0, K) \prod_i \sigma(y_i f_i)$ is non-Gaussian (no closed form).
MAP: $\hat{f} = \arg \min_f \sum_i \ln(1 + e^{-y_i f_i}) + \frac{1}{2} f^\top K^{-1} f$

Laplace approx: $\hat{\Sigma} = (K^{-1} + W)^{-1}$, W diagonal of log-likelihood second derivatives.
Predictive: $p(y_* = 1|y) = \int \sigma(f_*) p(f_*|y) df_*$
Decision boundary: $p(y_* = 1|y) = 0.5 \Leftrightarrow \mu_* = 0$.

Multi-Output GPs (MOGP)

Separable kernel: $K(x, x') = k(x, x')B$; stacked covariance $\tilde{K} = K_x \otimes B$.
 $\tilde{y} \sim \mathcal{N}(0, K_x \otimes B + \sigma^2 I)$

RKHS Reproducing Property: $\forall f \in \mathcal{H}_k$: $f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}$; $\|f\|_{\mathcal{H}_k}^2 = \alpha^\top K \alpha$ for $f = \sum_i \alpha_i k(x_i, \cdot)$.

LMC: $f_q(x) = \sum_{r=1}^R w_{qr} u_r(x)$, $u_r \sim \mathcal{GP}(0, k_r)$ i.i.d.
 $[K(x, x')]_{pq} = \sum_{r=1}^R w_{pr} w_{qr} k_r(x, x')$
Special case $R = 1$: $K(x, x') = k(x, x')(w w^\top)$ (separable kernel).

Missing Outputs: $\begin{bmatrix} f_{\text{obs}} \\ f_{\text{miss}} \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K_{\text{ooo}} & K_{\text{omm}} \\ K_{\text{moo}} & K_{\text{mmm}} \end{bmatrix}\right)$
 $f_{\text{miss}}|f_{\text{obs}} \sim \mathcal{N}(K_{\text{mo}} K_{\text{oo}}^{-1} f_{\text{obs}}, K_{\text{mm}} - K_{\text{mo}} K_{\text{oo}}^{-1} K_{\text{om}})$

Optimisation and SGD

Mini-batch gradient: $\nabla L_S(\theta) = \frac{1}{|S|} \sum_{i \in S} \nabla \ell(f(x_i; \theta), y_i)$
Unbiased: $\mathbb{E}[\nabla L_S(\theta)] = \nabla L(\theta)$
Variance: $\text{Var}[\nabla L_S(\theta)] = \frac{1}{|S|} \text{Var}[\nabla \ell]$ (shrinks with batch size)

Adam Bias Correction
 $m_k = \beta_1 m_{k-1} + (1 - \beta_1) g_k$, $m_0 = 0$. Unrolling:

$m_k = (1 - \beta_1) \sum_{s=1}^k \beta_1^{k-s} g_s$
 $\mathbb{E}[m_k] \approx (1 - \beta_1) \mu + \frac{1 - \beta_1^k}{1 - \beta_1} \mu = \mu(1 - \beta_1^k)$
Bias-corrected: $\hat{m}_k = m_k / (1 - \beta_1^k)$; likewise $\hat{v}_k = v_k / (1 - \beta_2^k)$. □

Hold-Out Validation is Unbiased

Condition on D : $\mathbb{E}[e(f(\tilde{x}_j; D), \tilde{y}_j)|D] = E^*(D)$. Average over m i.i.d. pairs and then over D : $\mathbb{E}[E_{\text{hold-out}}] = \mathbb{E}_D[E^*(D)] = E^*$. □

ROC-AUC = Pairwise Ranking Prob

$\text{AUC} = \frac{1}{n^+ n^-} \sum_i \sum_j [\mathbf{1}(s_i^+ > s_j^-) + \frac{1}{2} \mathbf{1}(s_i^+ = s_j^-)]$
Interpretation: probability a random positive is scored above a random negative. □

Cover-Hart Theorem As $n \rightarrow \infty$, $x^{(1)} \rightarrow x$ and $y^{(1)} \perp\!\!\!\perp y|x$.
Pointwise 1-NN error: $R_{1\text{-NN}}(x) = P(y \neq y^{(1)}|x) = 2\eta(x)(1 - \eta(x)) = 2R^*(x)(1 - R^*(x))$. Since $R^*(x) \leq \frac{1}{2}$: $R_{1\text{-NN}} \leq 2\mathbb{E}[R^*(x)] = 2R^*$. □

Stone's Theorem (stated) If $k \rightarrow \infty$ and $k/n \rightarrow 0$: $R_{k\text{-NN}} \rightarrow R^*$.
Excess risk $\lesssim \frac{1}{\sqrt{k}} + (\frac{k}{n})^{1/d}$; curse of dimensionality in the $1/d$ exponent.

Ensemble Methods: Bagging and Random Forests

Bagging (Bootstrap Aggregating)

Create B bootstrap samples $\mathcal{D}^{(b)} \sim \text{Bootstrap}(\mathcal{D})$. Train $f^{(b)} \leftarrow \mathcal{A}(\mathcal{D}^{(b)})$. Aggregate:
 $f_{\text{bag}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B f^{(b)}(\mathbf{x})$ (or majority vote).
Each bootstrap sample excludes any given point with prob $(1 - 1/n)^n \approx e^{-1} \approx 0.37$; $\approx 63\%$ of points appear at least once.

Bagging Variance Reduction (Derivation)

Assume $\mathbb{E}[f^{(b)}(\mathbf{x})] = \mu$, $\text{Var}(f^{(b)}(\mathbf{x})) = \sigma^2$, $\text{Corr}(f^{(b)}, f^{(c)}) = \rho$, $b \neq c$.
 $\text{Var}(\sum_{b=1}^B f^{(b)}) = B\sigma^2 + 2 \sum_{b < c} \rho \sigma^2 = B\sigma^2 + B(B - 1)\rho \sigma^2$

Divide by B^2 : $\text{Var}(f_{\text{bag}}(\mathbf{x})) = \frac{1 - \rho}{B} \sigma^2 + \rho \sigma^2$
 $\rho = 1$: no reduction (σ^2). $\rho = 0$: optimal (σ^2/B). As $B \rightarrow \infty$: floor $\rho \sigma^2$. □

Bias of Bagging: As $B \rightarrow \infty$, $f_{\text{bag}} \rightarrow \mathbb{E}[f^{(b)}|D]$. Bias $\approx$ bias of a single base model; larger B removes Monte Carlo noise only.

Out-of-Bag (OOB) Error: Let $\mathcal{B}_i = \{b : (x_i, y_i) \notin \mathcal{D}^{(b)}\}$.
 $f_{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} f^{(b)}(x_i)$, $E_{\text{OOB}} = \frac{1}{n} \sum_i e(y_i, f_{\text{OOB}}(x_i))$.
Provides a nearly free cross-validation estimate.

Random Forests: Reduce ρ by randomly subsampling q features at each split (instead of all p). Dominant predictors cannot monopolise splits; trees become more diverse. At each split: select $q \sim \sqrt{p}$ (classification) or $p/3$ (regression). Effect: $\sigma^2 \uparrow$, $\rho \downarrow$; net result often $\text{Var} \downarrow$.
 $f_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B f^{(b)}(\mathbf{x})$; trees grown fully (low bias).
Parallelisable; OOB error built-in. Bagging = variance reduction; Random Forests = variance reduction via decorrelated trees.

AdaBoost and Boosting

Ensemble: $h^{(B)}(\mathbf{x}) = \sum_{b=1}^B \alpha^{(b)} f^{(b)}(\mathbf{x})$, $h^{(0)} = 0$. Labels $y_i \in \{-1, +1\}$.
Loss: $L(h^{(b)}) = \sum_i \exp(-y_i h^{(b)}(x_i))$.

AdaBoost: Reweighting Derivation

$h^{(b)} = h^{(b-1)} + \alpha^{(b)} f^{(b)}$. Substitute and factorise:
 $L(f^{(b)}) = \sum_i \underbrace{\exp(-y_i h^{(b-1)}(x_i))}_{w_i^{(b)}} \exp(-y_i \alpha^{(b)} f^{(b)}(x_i))$

Since $y_i f^{(b)}(x_i) \in \{-1, +1\}$, define $\omega_c(f) = \sum_i w_i^{(b)} \mathbf{1}[y_i = f(x_i)]$, $\omega_e(f) = \sum_i w_i^{(b)} \mathbf{1}[y_i \neq f(x_i)]$:
 $L(\alpha, f) = e^{-\alpha \omega_c(f)} + e^{\alpha \omega_e(f)}$

Step 1 - Optimal $f^{(b)}$:

$\omega = \omega_c + \omega_e$ is independent of f ; since $e^\alpha - e^{-\alpha} > 0$:
 $f^{(b)} = \arg \min_f \sum_i w_i^{(b)} \mathbf{1}[y_i \neq f(x_i)]$ (weighted 0-1 loss) □

Step 2 - Optimal $\alpha^{(b)}$:

Fix $f^{(b)}$, differentiate $L(\alpha) = e^{-\alpha \omega_c} + e^{\alpha \omega_e}$:
 $\frac{dL}{d\alpha} = -e^{-\alpha \omega_c} \omega_c + e^{\alpha \omega_e} \omega_e = 0 \Rightarrow e^{2\alpha} \omega_c = \omega_c / \omega_e$
 $\alpha^{(b)} = \frac{1}{2} \ln \frac{\omega_c}{\omega_e} = \frac{1}{2} \ln \frac{1 - \varepsilon^{(b)}}{\varepsilon^{(b)}}$, $\varepsilon^{(b)} = \omega_e / \omega$

Weight Update:

$w_i^{(b+1)} = \exp(-y_i h^{(b)}(x_i)) = w_i^{(b)} \exp(-\alpha^{(b)} y_i f^{(b)}(x_i))$
Correct: $w \downarrow$; incorrect: $w \uparrow$. Requires $\varepsilon^{(b)} < 0.5$.

Bagging vs Boosting: Bagging: parallel, bootstrap, reduces variance (high-variance models). Boosting: sequential, reweighting, reduces bias (high-bias models). $h^{(b)}$ approximates increasingly rich function class.

AdaBoost Limitations: Exponential loss: misclassified points receive exponentially large weight. Problematic for outliers/mislabelled data. Cannot be parallelised.

Gradient Boosting

Generalises boosting to any differentiable loss $\ell(y, h(x))$. Optimise in function space: represent model by $h(\mathbf{X}) = (h(x_1), \dots, h(x_n))$.

Pseudo-Residuals (Negative Gradients):

$r_i^{(b)} = - \left. \frac{\partial \ell(y_i, z)}{\partial z} \right|_{z=h^{(b-1)}(x_i)}$

Algorithm: Init $h^{(0)} = \arg \min_c \sum_i \ell(y_i, c)$. For $b = 1, \dots, B$:

- Compute $r_i^{(b)}$ for all i .
- Fit regression tree $f^{(b)}$ to $\{(x_i, r_i^{(b)})\}$.
- Line search: $\alpha^{(b)} = \arg \min_\alpha \sum_i \ell(y_i, h^{(b-1)}(x_i) + \alpha f^{(b)}(x_i))$.
- Update: $h^{(b)}(x) = h^{(b-1)}(x) + \gamma \alpha^{(b)} f^{(b)}(x)$, $\gamma \in (0, 1]$.

Special Cases:

Squared loss: $\ell = (y - h)^2 \Rightarrow r_i^{(b)} = y_i - h^{(b-1)}(x_i)$ (residual fitting).
Exponential loss: $\ell = \exp(-yh) \Rightarrow$ recovers AdaBoost.
Logistic loss: $\ell = \log(1 + \exp(-yh)) \Rightarrow$ more robust (LogitBoost).

Logistic Pseudo-Residuals:

$\frac{\partial \ell}{\partial z} = \frac{-y}{1 + \exp(yz)} \Rightarrow r_i^{(b)} = \frac{y_i}{1 + \exp(y_i h^{(b-1)}(x_i))} = w_i^{(b)} y_i$
 $w_i^{(b)} = \frac{1}{1 + \exp(y_i h^{(b-1)}(x_i))} \in (0, 1]$: weights change *gradually* (vs AdaBoost's exponential blow-up).

Binary Classification Base Learner:

With $r_i^{(b)} = w_i^{(b)} y_i$, fitting f to $(x_i, r_i^{(b)})$ by squared loss $\equiv$ minimising $\sum_i w_i^{(b)} \mathbf{1}[y_i \neq f(x_i)]$, same as AdaBoost but different weights. Key: Boosting = gradient descent in function space.

Stochastic Gradient Boosting:

At each iteration b , sample $\mathcal{D}^{(b)} \subset \mathcal{D}$ without replacement ($\approx 50\text{--}80\%$ of data). Fit $f^{(b)}$ using only $\mathcal{D}^{(b)}$; update unchanged. Reduces correlation between learners $\Rightarrow$ lower variance. Combines bias reduction (boosting) with variance reduction (subsampling like bagging).

Optimisation: Adaptive Methods

Full Adam Update: Initialise $m_0 = v_0 = 0$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$.
 $m_k = \beta_1 m_{k-1} + (1 - \beta_1) g_k$ (1st moment)
 $v_k = \beta_2 v_{k-1} + (1 - \beta_2) g_k^2$ (2nd moment, elementwise)
 $\hat{m}_k = m_k / (1 - \beta_1^k)$, $\hat{v}_k = v_k / (1 - \beta_2^k)$
 $\theta_{k+1} = \theta_k - \eta \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \varepsilon}$
Intuition: adapts per-parameter learning rate; $\sqrt{\hat{v}_k}$ scales by RMS of past gradients; $\hat{m}_k$ provides momentum.

AdaGrad: Accumulate squared gradients $V_k = \sum_{s=1}^k g_s^2$ (elementwise).

$\theta_{k+1} = \theta_k - \frac{\eta}{\sqrt{V_k} + \varepsilon} \odot g_k$

Monotonically increasing $V_k \Rightarrow$ learning rate only decreases. Good for sparse gradients; may stop learning too early for dense.

RMSProp: Exponential moving average of squared gradients.

$v_k = \beta v_{k-1} + (1 - \beta) g_k^2$
 $\theta_{k+1} = \theta_k - \frac{\eta}{\sqrt{v_k} + \varepsilon} \odot g_k$

Fixes AdaGrad's diminishing LR by forgetting old gradients. Adam = RMSProp + momentum + bias correction.

Performance Evaluation Metrics

Confusion Matrix Components: TP, FP, TN, FN from predicted vs actual class.

Accuracy = $\frac{TP + TN}{TP + FP + TN + FN}$

Precision = $\frac{TP}{TP + FP}$, Recall = Sensitivity = $\frac{TP}{TP + FN}$

Specificity = $\frac{TN + FP}{TN + FP + FN}$, FPR = $\frac{FP}{FP + TN} = 1 - \text{Spec}$.

ROC plots TPR (Recall) vs FPR across thresholds.

F-Scores:

$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$ (harmonic mean)

$F_\beta = \frac{(1 + \beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$; $\beta > 1$ weights recall, $\beta < 1$ weights precision.

Macro vs Micro Averaging (Multiclass, C classes):

Macro: compute metric per class, then average (treats all classes equally regardless of size):

Macro-P = $\frac{1}{C} \sum_{c=1}^C \text{Precision}_c$

Micro: aggregate TP, FP, FN across all classes, then compute (weights by class frequency):

Micro-P = $\frac{\sum_c TP_c}{\sum_c (TP_c + FP_c)}$

Macro is sensitive to rare class performance; micro is dominated by large classes.

Regularisation: Lasso and Ridge

Ridge (ℓ_2): $\hat{\theta} = \arg \min_\theta \frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_1^2 = (X^\top X + n\lambda I)^{-1} X^\top y$. Shrinks all coefficients; closed-form solution; always invertible.

Lasso (ℓ_1): $\hat{\theta} = \arg \min_\theta \frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_1$. Promotes sparsity (many $\hat{\theta}_j = 0$) due to non-smooth ℓ_1 ball corners. No closed form; solved via coordinate descent or subgradient methods.

Geometric intuition: Ridge constraint is a ball $\|\theta\|_2^2 \leq t$ (smooth); least-squares contours touch it at non-sparse points. Lasso constraint is a diamond $\|\theta\|_1 \leq t$ (corners on axes); contours likely touch at corners $\Rightarrow$ sparsity.

Additional Kernels

Kernel-Induced Distance:

$\text{dist}_k^2(x, x') = k(x, x) + k(x', x') - 2k(x, x')$
(Squared Euclidean distance in the RKHS feature space ϕ .)

RBF/Gaussian Kernel: $k(x, x') = \exp\left(-\frac{\|x - x'\|_2^2}{2\ell^2}\right)$

Smooth, infinitely differentiable; infinite-dimensional feature map.

Laplacian Kernel: $k(x, x') = \exp\left(-\frac{\|x - x'\|_1}{\sigma}\right)$

Uses ℓ_1 distance; less smooth than RBF; more robust to outliers.

Sigmoid Kernel: $k(x, x') = \tanh(\gamma x^\top x' + c)$
Not always PSD (valid only for certain γ, c); related to neural networks.

Polynomial Kernel (general): $k(x, x') = (x^\top x' + c)^d$, $c \geq 0$, $d \in \mathbb{Z}^+$. Always PSD; explicit feature map is degree- d monomials.

High-Dimensional Geometry

Curse of Dimensionality (Volume): To capture fraction p of data uniformly distributed in $[0, 1]^d$, a hypercube side length ℓ satisfies:

$\ell^d = p \Rightarrow \ell = p^{1/d}$
For $d = 10$, $p = 0.01$: $\ell = 0.01^{0.1} \approx 0.63$. Even a tiny fraction requires most of the range in each dimension.

Regression Loss Functions

Huber Loss: Interpolates between squared and absolute loss.

$\ell_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \leq \delta \\ \delta(|r| - \frac{1}{2}\delta) & |r| > \delta \end{cases}$
Quadratic for small residuals (smooth, efficient); linear for large (robust to outliers). Bayes-optimal predictor: conditional mean (for symmetric noise). Requires tuning δ .

Poisson Distribution

$p(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!}$, $y \in \{0, 1, 2, \dots\}$, $\lambda > 0$.

$\mathbb{E}[y] = \lambda$, $\text{Var}(y) = \lambda$. Mean = Variance.
NLL: $-\ln p(y; \lambda) = \lambda - y \ln \lambda + \ln y!$. With $\lambda_i = \exp(\theta^\top x_i)$:

NLL = $\sum_i (e^{\theta^\top x_i} - y_i \theta^\top x_i) + \text{const}$
Link function: $\phi(\mu) = \log \mu$ (log link); keeps rate $\lambda > 0$ for all θ .